Streetmap UK MapLibre Style

https://gemini.google.com/app/e05cce7a112fb535

User prompt: Can you create a MapLibre style based on the map style from streetmap.co.uk

Response: The classic look of **Streetmap.co.uk** is based on traditional British cartography, particularly the iconic **Ordnance Survey (OS) Landranger / Street Atlas** aesthetic:

- **Background / Landuse:** Clean off-white/cream background (`#FFFDF7` / `#F8F6EB`).
- **Parks / Open Green:** Muted pale green (`#D8E8C5` / `#C7E1B8`).
- **Water:** Distinct cyan/pale blue (`#98D2EB`).
- **Buildings / Urban fill:** Very pale greyish-tan (`#EBE6DF`).
- **Motorways:** Distinct British motorway blue (`#0079C1` casing with white or vivid blue).
- **Primary / 'A' Roads:** OS emerald green fill (`#009A44`) with white dashed/solid centerline or casing.
- **Secondary / 'B' Roads:** OS mustard/amber-orange (`#EE8200` or `#F7941D`).
- **Local / Residential Streets:** Clean white casings with crisp dark slate/black boundaries (`#FFFFFF` with `#707070` or `#404040` casing).
- **Railways:** Traditional black line with perpendicular cross-ties/dashes.

Here is a ready-to-use **MapLibre GL style JSON** tailored to an OpenMapTiles / Protomaps vector tile schema (compatible with free services like MapTiler, Stadia Maps, or self-hosted PMTiles):

```
{
  "version": 8,
  "name": "Streetmap UK Classic",
  "glyphs": "https://demotiles.maplibre.org/font/{fontstack}/{range}.pbf",
  "sources": {
    "openmaptiles": {
      "type": "vector",
      "url": "https://demotiles.maplibre.org/tiles/tiles.json"
    }
  },
  "layers": [
    {
      "id": "background",
      "type": "background",
      "paint": {
        "background-color": "#fcfbf7"
      }
    },
    {
      "id": "landuse_park",
      "type": "fill",
      "source": "openmaptiles",
      "source-layer": "park",
      "paint": {
        "fill-color": "#d9eec7",
        "fill-opacity": 0.8
      }
    },
    {
      "id": "water",
      "type": "fill",
      "source": "openmaptiles",
      "source-layer": "water",
      "paint": {
        "fill-color": "#9ed5ea",
        "fill-outline-color": "#6eaec8"
      }
    },
    {
      "id": "building",
      "type": "fill",
      "source": "openmaptiles",
      "source-layer": "building",
      "paint": {
        "fill-color": "#eae4db",
        "fill-outline-color": "#cdc5b9"
      }
    },
    {
      "id": "tunnel_casing",
      "type": "line",
      "source": "openmaptiles",
      "source-layer": "transportation",
      "filter": ["==", "brunnel", "tunnel"],
      "paint": {
        "line-color": "#a0a0a0",
        "line-width": 4,
        "line-dasharray": [2, 2]
      }
    },
    {
      "id": "road_residential_casing",
      "type": "line",
      "source": "openmaptiles",
      "source-layer": "transportation",
      "filter": ["all", ["==", "$type", "LineString"], ["in", "class", "minor", "service", "residential"]],
      "paint": {
        "line-color": "#9e9e9e",
        "line-width": ["interpolate", ["linear"], ["zoom"], 12, 1.5, 16, 7]
      }
    },
    {
      "id": "road_residential_fill",
      "type": "line",
      "source": "openmaptiles",
      "source-layer": "transportation",
      "filter": ["all", ["==", "$type", "LineString"], ["in", "class", "minor", "service", "residential"]],
      "paint": {
        "line-color": "#ffffff",
```

```
    "line-width": ["interpolate", ["linear"], ["zoom"], 12, 0.8, 16, 5]
  }
},
{
  "id": "road_secondary_casing",
  "type": "line",
  "source": "openmaptiles",
  "source-layer": "transportation",
  "filter": ["all", ["==", "$type", "LineString"], ["in", "class", "secondary", "tertiary"]],
  "paint": {
    "line-color": "#8c4400",
    "line-width": ["interpolate", ["linear"], ["zoom"], 10, 2, 16, 9]
  }
},
{
  "id": "road_secondary_fill",
  "type": "line",
  "source": "openmaptiles",
  "source-layer": "transportation",
  "filter": ["all", ["==", "$type", "LineString"], ["in", "class", "secondary", "tertiary"]],
  "paint": {
    "line-color": "#f59c1a",
    "line-width": ["interpolate", ["linear"], ["zoom"], 10, 1.2, 16, 7]
  }
},
{
  "id": "road_primary_casing",
  "type": "line",
  "source": "openmaptiles",
  "source-layer": "transportation",
  "filter": ["all", ["==", "$type", "LineString"], ["==", "class", "primary"]],
  "paint": {
    "line-color": "#004d25",
    "line-width": ["interpolate", ["linear"], ["zoom"], 8, 2, 16, 10]
  }
},
{
  "id": "road_primary_fill",
  "type": "line",
  "source": "openmaptiles",
  "source-layer": "transportation",
  "filter": ["all", ["==", "$type", "LineString"], ["==", "class", "primary"]],
  "paint": {
    "line-color": "#009a44",
    "line-width": ["interpolate", ["linear"], ["zoom"], 8, 1.2, 16, 8]
  }
},
{
  "id": "road_motorway_casing",
  "type": "line",
  "source": "openmaptiles",
  "source-layer": "transportation",
  "filter": ["all", ["==", "$type", "LineString"], ["==", "class", "motorway"]],
  "paint": {
    "line-color": "#003b6d",
    "line-width": ["interpolate", ["linear"], ["zoom"], 6, 2.5, 16, 12]
  }
},
{
  "id": "road_motorway_fill",
  "type": "line",
  "source": "openmaptiles",
  "source-layer": "transportation",
  "filter": ["all", ["==", "$type", "LineString"], ["==", "class", "motorway"]],
  "paint": {
    "line-color": "#0079c1",
    "line-width": ["interpolate", ["linear"], ["zoom"], 6, 1.5, 16, 10]
  }
},
{
  "id": "railway_base",
  "type": "line",
  "source": "openmaptiles",
  "source-layer": "transportation",
  "filter": ["==", "class", "rail"],
  "paint": {
    "line-color": "#444444",
    "line-width": 2
  }
},
{
  "id": "railway_ticks",
  "type": "line",
  "source": "openmaptiles",
  "source-layer": "transportation",
  "filter": ["==", "class", "rail"],
  "paint": {
    "line-color": "#ffffff",
    "line-width": 1.5,
    "line-dasharray": [1.5, 1.5]
  }
},
{
  "id": "road_label",
  "type": "symbol",
  "source": "openmaptiles",
  "source-layer": "transportation_name",
  "layout": {
    "symbol-placement": "line",
    "text-field": "{name}",
    "text-font": ["Noto Sans Regular"],
    "text-size": 11,
    "text-letter-spacing": 0.05
  },
  "paint": {
    "text-color": "#1b1b1b",
```

```
        "text-halo-color": "#ffffff",
        "text-halo-width": 1.5
      }
    },
    {
      "id": "place_label_town",
      "type": "symbol",
      "source": "openmaptiles",
      "source-layer": "place",
      "filter": ["in", "class", "town", "city"],
      "layout": {
        "text-field": "{name}",
        "text-font": ["Noto Sans Bold"],
        "text-size": ["interpolate", ["linear"], ["zoom"], 8, 12, 14, 18],
        "text-transform": "uppercase"
      },
      "paint": {
        "text-color": "#222222",
        "text-halo-color": "#ffffff",
        "text-halo-width": 2
      }
    }
  ]
}
```

Implementing in MapLibre GL JS

To load this style, point your MapLibre instance to the JSON file or pass it directly:

```
const map = new maplibregl.Map({
  container: 'map',
  style: 'streetmap-style.json', // or inline style object
  center: [-0.1278, 51.5074], // London
  zoom: 14
});

// Add standard navigation controls
map.addControl(new maplibregl.NavigationControl(), 'top-right');
```

Tips for Perfecting the OS / Streetmap Look:

1. **Font Choice:** Streetmap/OS cartography heavily uses clean grotesque sans-serifs (like **Gill Sans**, **Transport**, or **DIN 1451**). Setting `text-font` to a Transport or Gill Sans fontstack replicates British road signs and OS styling accurately.
2. **Road Reference Numbers:** Add a dedicated symbol layer for `ref` fields (e.g., `M25`, `A4`, `B3124`) styled with rounded pill-boxes/shields using UK signpost color conventions (White-on-Blue for motorways, Yellow-on-Green for primary A-roads).

User prompt: Is it possible to recreate significant signage?

Response: Yes. You can recreate significant British road signage in MapLibre by combining **vector tile attributes** (`ref`, `network`, `shield`) with MapLibre's **shield/icon rendering** and **formatted text formatting**.

In the UK, the Traffic Signs Regulations and General Directions (TSRGD) define four distinct sign badge styles:

Route Type	Sign Background	Text Color	Format / Style
Motorways (`M` / `(M)`)	Deep Blue (`#0079C1`)	White (`#FFFFFF`)	Rounded rectangle, Transport Heavy / Motorway font
Primary Routes (`A`)	Dark Green (`#00703C`)	Yellow (`#FFD200`)	Rounded rectangle, green ground
Non-Primary (`A` / `B`)	White (`#FFFFFF`)	Black (`#111111`)	Black border casing
Motorway Junctions	Deep Blue (`#0079C1`)	White (`#FFFFFF`)	Black/white patch with junction "J" or number

Implementation Method 1: Vector Sprite Shields (Recommended)

This produces authentic rounded rectangular badges behind the text.

1. **Add badge icons to your sprite sheet:** Add small stretchable SVG images (9-slice shields) named `shield_uk_motorway`, `shield_uk_primary`, and `shield_uk_nonprimary`.
2. **Add Symbol Layers to the MapLibre Style:**

```
[
  {
    "id": "ref_motorway_badge",
    "type": "symbol",
    "source": "openmaptiles",
    "source-layer": "transportation_name",
    "filter": [
      "all",
      ["==", "class", "motorway"],
      ["has", "ref"]
    ],
    "layout": {
      "icon-image": "shield_uk_motorway",
      "icon-text-fit": "both",
      "icon-text-fit-padding": [2, 5, 2, 5],
      "text-field": "{ref}",
      "text-font": ["Noto Sans Bold"],
      "text-size": 10,
      "symbol-placement": "line",
      "text-rotation-alignment": "viewport",
      "icon-rotation-alignment": "viewport"
    },
    "paint": {
      "text-color": "#ffffff"
    }
  },
  {
    "id": "ref_primary_a_road_badge",
    "type": "symbol",
    "source": "openmaptiles",
    "source-layer": "transportation_name",
    "filter": [
```

```
      "all",
      ["==", "class", "primary"],
      ["has", "ref"]
    ],
    "layout": {
      "icon-image": "shield_uk_primary",
      "icon-text-fit": "both",
      "icon-text-fit-padding": [2, 4, 2, 4],
      "text-field": "{ref}",
      "text-font": ["Noto Sans Bold"],
      "text-size": 10,
      "symbol-placement": "line",
      "text-rotation-alignment": "viewport",
      "icon-rotation-alignment": "viewport"
    },
    "paint": {
      "text-color": "#ffd200"
    }
  }
]
```

`icon-text-fit: "both"` automatically stretches the background pill to fit any reference code (e.g., `M4`, `M25`, or `A369`).

Implementation Method 2: Dynamic Canvas Badges (No Sprite Sheet Required)

If you don't want to build custom sprites, register dynamic UK sign patches programmatically using MapLibre's `addImage` API:

```
// Function to generate a UK road sign pill dynamically
function createUkShield(fillColor, strokeColor) {
  const canvas = document.createElement('canvas');
  canvas.width = 64;
  canvas.height = 32;
  const ctx = canvas.getContext('2d');

  const r = 6;
  const w = canvas.width - 4;
  const h = canvas.height - 4;
  const x = 2;
  const y = 2;

  ctx.beginPath();
  ctx.moveTo(x + r, y);
  ctx.arcTo(x + w, y, x + w, y + h, r);
  ctx.arcTo(x + w, y + h, x, y + h, r);
  ctx.arcTo(x, y + h, x, y, r);
  ctx.arcTo(x, y, x + w, y, r);
  ctx.closePath();

  ctx.fillStyle = fillColor;
  ctx.fill();

  if (strokeColor) {
    ctx.lineWidth = 2;
    ctx.strokeStyle = strokeColor;
    ctx.stroke();
  }

  return ctx.getImageData(0, 0, canvas.width, canvas.height);
}

// Register shields once the map style loads
map.on('load', () => {
  // Motorway: Blue
  map.addImage('shield_uk_motorway', createUkShield('#0079C1'), {
    content: [8, 6, 56, 26],
    stretchX: [[12, 52]],
    stretchY: [[8, 24]]
  });

  // Primary Route (A-road): Green
  map.addImage('shield_uk_primary', createUkShield('#00703C'), {
    content: [8, 6, 56, 26],
    stretchX: [[12, 52]],
    stretchY: [[8, 24]]
  });

  // Non-Primary Route (B-road / local A): White with black border
  map.addImage('shield_uk_nonprimary', createUkShield('#FFFFFF', '#111111'), {
    content: [8, 6, 56, 26],
    stretchX: [[12, 52]],
    stretchY: [[8, 24]]
  });
});
```

Essential Typography Nuances

- **Font Family:** True UK road signs use the **Transport** typeface (designed by Jock Kinneir and Margaret Calvert) for non-motorway roads, and **Motorway** typeface for motorway route numbers. Free open-source alternatives like **Transport New** or **Overpass** can be compiled into PBF glyphs.
- **Junction Numbers:** In OpenStreetMap vector tiles, junctions are tagged under the `transportation_name` or `highway` layer with `highway=motorway_junction`. You can render them with a small black square badge displaying the junction number (e.g., `[ 19 ]`).